H. Can You Cut the Ropes?

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given n ropes with positive integer lengths. You may cut the ropes into smaller pieces (each with positive integer length).

Your task is to obtain at least k pieces such that every piece has the same positive integer length. Any leftover part after cutting may be discarded.

Find the maximum possible length of each piece. If it is not possible to obtain at least k pieces of any positive integer length, print -1.

Input

The first line contains two integers n $(1 \le n \le 2 \times 10^5)$ and k $(1 \le k \le 10^{18})$, the number of ropes and the required number of pieces respectively.

The next line contains n integers $l_1, l_2, \ldots, l_n$ $(1 \le l_1, \ldots, l_n \le 10^9)$, the lengths of the ropes.

Output

Print one integer: the maximum possible length of each piece.

If it is not possible, print -1.

Examples

input

```
4 8
2 6 4 4
```

output

```
2
```

input

```
5 1000
1 2 3 4 5
```

output

```
-1
```

Note

Brief Explanation for the First Sample: All of our final ropes have to be of the same length.

- Suppose that we want our final ropes to have length 1. We can get $2 + 6 + 4 + 4 = 12$ pieces of ropes of length 1. We can
  - cut the first rope into $\frac{2}{1} = 2$ pieces of length 1,
  - cut the second rope into $\frac{6}{1} = 6$ pieces of length 1,
  - cut the third rope into $\frac{4}{1} = 4$ pieces of length 1, and
  - cut the fourth rope into $\frac{4}{1} = 4$ pieces of length 1.
- Suppose that we want to our final ropes to have length 2. We can get $1 + 3 + 2 + 2 = 8$ pieces of ropes of length 2. We can

- cut the first rope into $\frac{2}{2} = 1$ piece of length 2,
- cut the second rope into $\frac{6}{2} = 3$ pieces of length 2,
- cut the third rope into $\frac{4}{2} = 2$ pieces of length 2, and
- cut the fourth rope into $\frac{4}{2} = 2$ pieces of length 2.

- Suppose that we want our final ropes to have length 3. We can get $0 + 2 + 1 + 1 = 4$ pieces of ropes of length 3.
  - $\frac{2}{3} = 0.667$, so we can get 0 pieces of length 3;
  - $\frac{6}{3} = 2$, so we can get 2 pieces of length 3;
  - $\frac{4}{3} = 1.33$, so we can get 1 piece of length 3; and
  - $\frac{4}{3} = 1.33$, so we can get 1 piece of length 3.

To summarize, we can get 12 pieces of ropes of length 1, 8 pieces of ropes of length 2, and 4 pieces of ropes of length 3. Since we need at least $k = 8$ pieces, the answer can not be 3. It also can not be 1, as we want the maximum length. So the answer is 2.